Homework 4 – Theory

1

(a)

The probabilities must add to 1 and be non-negative:

$$\theta_1 + 2\theta_1 + \theta_2 = 1.$$

$$\boxed{3\theta_1 + \theta_2 = 1, \qquad \theta_1 \geq 0, \quad \theta_2 \geq 0}$$

(b)

The observations are independent, so

$$\begin{aligned} P(D \mid \theta) &= \prod_{i=1}^{n} P(x^{(i)} \mid \theta) \\ &= \theta_1^{s_1} (2\theta_1)^{s_2} \theta_2^{s_3} \\ &= 2^{s_2} \theta_1^{s_1+s_2} \theta_2^{s_3}. \end{aligned}$$

Taking the log,

$$\boxed{\log P(D \mid \theta) = s_2 \log 2 + (s_1 + s_2) \log \theta_1 + s_3 \log \theta_2}$$

(c)

Using $\theta_2 = 1 - 3\theta_1$,

$$\ell(\theta_1) = s_2 \log 2 + (s_1 + s_2) \log \theta_1 + s_3 \log(1 - 3\theta_1).$$

Set the derivative to zero:

$$\frac{d\ell}{d\theta_1} = \frac{s_1 + s_2}{\theta_1} - \frac{3s_3}{1 - 3\theta_1} = 0.$$

Solving,

$$(s_1 + s_2)(1 - 3\theta_1) = 3s_3\theta_1,$$

$$s_1 + s_2 = 3(s_1 + s_2 + s_3)\theta_1 = 3n\theta_1.$$

Then,

$$\hat{\theta}_1 = \frac{s_1 + s_2}{3n}, \qquad \hat{\theta}_2 = 1 - 3\hat{\theta}_1 = \frac{s_3}{n}.$$

The log-likelihood is concave, so this is a maximum. The formula also holds at the endpoints.

$$\boxed{\hat{\theta} = \left[\frac{s_1 + s_2}{3n}, \frac{s_3}{n}\right]}$$

2

The likelihood is

$$\begin{aligned} L(\beta) &= \prod_{i=1}^{n} \frac{1}{\beta} e^{-x^{(i)}/\beta} \\ &= \beta^{-n} \exp\left(-\frac{1}{\beta}\sum_{i=1}^{n} x^{(i)}\right). \end{aligned}$$

Taking the log,

$$\ell(\beta) = -n\log\beta - \frac{1}{\beta}\sum_{i=1}^{n} x^{(i)}.$$

Set the derivative to zero:

$$\frac{d\ell}{d\beta} = -\frac{n}{\beta} + \frac{1}{\beta^2}\sum_{i=1}^{n} x^{(i)} = 0.$$

Multiplying by β^2,

$$-n\beta + \sum_{i=1}^{n} x^{(i)} = 0.$$

For $\bar{x} > 0$, the derivative changes from positive to negative at $\beta = \bar{x}$, so this is the maximum.

$$\boxed{\hat{\beta} = \frac{1}{n}\sum_{i=1}^{n} x^{(i)} = \bar{x}}$$

3

(a)

There are 5 defective items out of 30:

$$p(D \mid \theta) = \binom{30}{5}\theta^5(1-\theta)^{25}.$$

The prior is uniform, so $p(\theta) = 1$ on $[0, 1]$.

Using Bayes' rule,

$$p(\theta \mid D) \propto p(D \mid \theta)p(\theta).$$

Removing the constant,

$$\boxed{p(\theta \mid D) \propto \theta^5(1-\theta)^{25}, \qquad 0 \leq \theta \leq 1}$$

This is a Beta$(6, 26)$ distribution.

(b)

Given $x^{(i)} \sim \mathcal{N}(\mu, 1)$ and $\mu \sim \mathcal{N}(0, 1)$,

$$p(\mu \mid D) \propto p(D \mid \mu)p(\mu).$$

Using the likelihood and prior,

$$p(\mu \mid D) \propto \exp\left(-\frac{1}{2}\left[\sum_{i=1}^{n}(x^{(i)}-\mu)^2 + \mu^2\right]\right).$$

Expanding,

$$\begin{aligned}&\sum_{i=1}^{n}(x^{(i)}-\mu)^2 + \mu^2 \\ &= (n+1)\mu^2 - 2\mu\sum_{i=1}^{n}x^{(i)} + \sum_{i=1}^{n}(x^{(i)})^2.\end{aligned}$$

Completing the square and removing constants,

$$p(\mu \mid D) \propto \exp\left(-\frac{n+1}{2}\left(\mu - \frac{\sum_{i=1}^{n}x^{(i)}}{n+1}\right)^2\right).$$

The mean and variance are

$$\text{Mean} = \frac{\sum_{i=1}^{n}x^{(i)}}{n+1}, \qquad \text{Variance} = \frac{1}{n+1}.$$

Thus,

$$\boxed{\mu \mid D \sim \mathcal{N}\left(\frac{\sum_{i=1}^{n}x^{(i)}}{n+1}, \frac{1}{n+1}\right)}$$